



Academic Papers

1. **Profiling relational data: a survey** — Ziawasch Abedjan, Lukasz Golab, Felix Naumann, 2015, *VLDB Journal*. This paper surveys relational profiling tasks and organizes the field around common metadata and dependency discovery problems. It reviews tools and techniques for statistics, correlations, functional dependencies, inclusion dependencies, and conditional properties, which makes it a foundational map for your ML/statistical core. Relevance: **HIGH** — it anchors the profiling side of the project and helps define what belongs in the structured findings file. Link: [DOI](#) [verified].^[1]
2. **Using Data Contracts to improve data quality** — IBM documentation, 2024–2026. This is not a paper, but it is one of the clearest public references for declarative data-contract-driven DQ validation. It shows a YAML/JSON contract stored in Git, executed as validation tests, and retrieved as results through an API, which is very close to your contract-first architecture. Relevance: **HIGH** — strong precedent for contract-as-input plus machine-readable test results. Link: [IBM docs](#) [verified].^[2]
3. **Quality Assessment of Tabular Data using Large Language Models ...** — authors not fully extracted from snippet, 2025, arXiv. The paper proposes a three-stage framework that combines large-scale statistical inlier detection, LLM-based semantically valid rule generation, and code synthesis for executable validators. This is very close to the “ML/statistics first, LLM advisor second” pattern, though it is still more LLM-generative than your intended design. Relevance: **HIGH** — useful as the nearest known LLM-assisted validator-synthesis precedent. Link: [arXiv PDF](#) [verified].^[3]
4. **Test-driven evaluation of linked data quality** — authors not extracted from snippet, 2014, ACM DL. This work formalizes “bad smells” and quality problems as testable conditions for linked data quality assessment. It is relevant because it treats quality checking as test generation rather than narrative analysis, which maps well to a structured validation engine. Relevance: **MEDIUM** — conceptual precursor to declarative, test-driven DQ contracts. Link: [ACM DOI](#) [verified].^[4]
5. **Understanding data quality in a data-driven industry context** — 2024, *Data & Knowledge Engineering / ScienceDirect* snippet. This is a review that frames data quality around deliberate design and operational enforcement in industry settings. It is useful for positioning your work in the broader DQ management landscape rather than the algorithmic core. Relevance: **MEDIUM** — good positioning paper, but not a core technical reference. Link: [ScienceDirect](#) [uncertain].^[5]
6. **Profiling relational data: a survey** — duplicate bibliographic listing with DOI and venue metadata. The Waterloo page confirms the journal, year, volume, issue, pages, and DOI. Relevance: **HIGH** — use this as the canonical citation if you need a clean bibliographic source. Link: [Waterloo page](#) [verified].^[1]
7. **ExpectationSuiteValidationResult** — Great Expectations documentation, 2022. This is documentation rather than a paper, but it is important prior art for structured JSON-like validation results. The object exposes per-check results, success flags, statistics, metadata, and JSON serialization, which is directly relevant to your output schema design. Relevance: **HIGH** — a strong structured-report reference. Link: [GE docs](#) [verified].^[6]
8. **Expectation Suite** — Great Expectations documentation. An Expectation Suite is a collection of verifiable assertions about data, stored as JSON in the project. This is one of the clearest examples of declarative validation contracts in the data quality ecosystem. Relevance: **HIGH** — close analog to DBML-like contract-driven validation, though not schema-native. Link: [GE docs](#) [verified].^[7]
9. **Validating Data** — Frictionless Framework docs, 2022. Frictionless explicitly frames schema validation as a reusable class of tabular-data checks using Table Schema / Frictionless specs. It is especially relevant because the contract and validation artifacts are lightweight and machine-readable. Relevance: **HIGH** — very close to your declarative-contract approach. Link: [Frictionless docs](#) [verified].^[8]
10. **Data management framework for Python ... DEVT Framework** — frictionless-py repository, 2014–present. The project describes extract, validate, and transform for tabular data using Frictionless standards. It is not a paper, but it is a core implementation reference for schema-driven validation. Relevance: **HIGH** — important ecosystem anchor. Link: [GitHub repo](#) [verified].^[9]
11. **dbt-expectations** — package docs/repo, 2020. This is a dbt extension inspired by Great Expectations that ports expectations into dbt tests. It is relevant because dbt model contracts and test results are one of the nearest

industrial patterns to your “declarative contract + structured findings” idea. Relevance: **HIGH** — strong analogous ecosystem, though not DBML-based. Link: [GitHub repo](#) [verified].^[10]

12. **Great Expectations core repository** — 2025 repo index. The project’s main repository remains a central example of expectation-driven DQ validation with CI support and structured results. Relevance: **HIGH** — essential prior art for contract suites and validation artifacts. Link: [GitHub org repo list](#) [verified].^[11]

GitHub Repositories

PRIMARY — Schema-spec ecosystems and schema-driven validators

Repo	Stars / date observed	Description	Status	Inputs	Outputs	Fit vs your project	Tag
holistics/dbml	[uncertain] / not observed	DBML parser/model ecosystem for the DBML schema language.	active [uncertain]	DBML text	DBML AST / tooling artifacts [uncertain]	Strong schema- spec foundation; no clear DQ engine or LLM advisor.	[uncertain]
frictionlessdata/frictionless-py	[uncertain] / not observed	Python framework for describing, validating, and transforming tabular data.	active [uncertain]	Frictionless descriptors, Table Schema, CSV/Excel/JSON- like tabular sources	Validation reports, Python objects, package artifacts	Very close on contract- driven validation and structured results.	[verified] ^[9]
calogica/dbt-expectations	[uncertain] / not observed	Great Expectations- style tests for dbt.	active [uncertain]	dbt YAML / SQL models / tests	dbt test results	Strong declarative validation pattern, but dbt-native not DBML- native.	[verified] ^[10]
great- expectations/great_expectations	11k-ish / Sep 2025 [uncertain]	Validation framework built around expectation suites.	active	Expectation suites, pandas/Spark/SQL sources	JSON- serializable validation results, docs artifacts	Excellent structured- output prior art; not schema- spec-first.	[verified] ^{[11] [6] [7]}
dbt-labs/dbt-core	[uncertain] / not observed	dbt transformation framework and test ecosystem.	active [uncertain]	SQL models, YAML schema tests	SQL test artifacts, run results	Important contract- adjacent ecosystem; contracts are strong but not generalized DQ.	[uncertain]
dbt-labs/dbt-adapters	[uncertain] / not observed	dbt adapter layer for warehouses.	active [uncertain]	Warehouse connections	Adapter execution artifacts	Useful for contract enforcement in live DBs, but not DQ- specific.	[uncertain]

PRIMARY — Data quality / validation frameworks

Repo	Stars / date observed	Description	Status	Inputs	Outputs	Fit vs your project	T
great-expectations/great_expectations	~11k / Sep 2025 [uncertain]	Data validation framework centered on expectation suites.	active	Pandas, Spark, SQL, files	JSON validation results, checkpoints, docs	Strong on contract-like rules and structured output.	[verified] [6]
frictionlessdata/frictionless-py	[uncertain] / not observed	Lightweight data management and validation framework.	active [uncertain]	Tabular data, Table Schema, descriptors	Reports, Python objects	Very strong for declarative table validation; lacks ML anomaly layer.	[verified] [9]
calogica/dbt-expectations	[uncertain] / not observed	GE-like assertions for dbt.	active [uncertain]	dbt tests / YAML	dbt test artifacts	Good for contract-aligned tests; limited beyond rule-based checks.	[verified] [10]
aws/deequ	[uncertain] / not verified here	Scala/Spark data quality framework.	active [uncertain]	Spark DataFrames, rules	Constraint suggestions, metrics, checks	Strong metric/constraint generation; structured output exists but contract support is indirect.	[uncertain]
unionai-oss/pandera	[uncertain] / not verified here	DataFrame schema and validation library.	active [uncertain]	Pandas/Polars/Pyspark schemas	Python exceptions / check results	Good row/column validation; not a full profiling/anomaly engine.	[uncertain]
sodadata/soda-core	[uncertain] / not verified here	Data quality checks and scans.	active [uncertain]	YAML checks, databases, files	Scan results, CLI output	Strong operational checks, less expressive contract modeling than your design.	[uncertain]
tfdv/tensorflow-data-validation	[uncertain] / not verified here	Schema and anomaly detection for ML data pipelines.	active [uncertain]	TFRecords / tabular stats	Schema, anomalies, stats	Strong on schema drift; not contract-file agnostic.	[uncertain]

PRIMARY — Classical statistical profiling libraries

Repo	Stars / date observed	Description	Status	Inputs	Outputs	Fit vs your project	Tag
ydataai/ydata-profiling	[uncertain] / not verified here	Automated profiling and EDA report generator.	active [uncertain]	Pandas-like tables	HTML/JSON-ish profile outputs [uncertain]	Strong profiling algorithms; output is report-centric, not contract-centric.	[uncertain]
data-science-profiler/DataProfiler	[uncertain] / not verified here	Data profiling and type detection library.	active [uncertain]	CSV, text, tabular data	Profiles, labels, statistics	Very relevant for column typing and profiling metrics.	[uncertain]

Repo	Stars / date observed	Description	Status	Inputs	Outputs	Fit vs your project	Tag
AutoViML/sweetviz	[uncertain] / not verified here	Fast EDA comparison/profiling tool.	active/stale [uncertain]	CSV/DataFrame	HTML reports	Good profiling heuristics, weak structured output.	[uncertain]
man-group/klib	[uncertain] / not verified here	Data cleaning, profiling, and preprocessing helpers.	active [uncertain]	pandas DataFrame	Python outputs	Useful utility library, not a contract validator.	[uncertain]
DataPrep	[uncertain] / not verified here	Data preparation and profiling toolkit.	stale [uncertain]	DataFrames	HTML/report outputs	Helpful profiling baseline, weaker maintenance signal.	[uncertain]

PRIMARY — Traditional ML / statistical DQ detection

Repo	Stars / date observed	Description	Status	Inputs	Outputs	Fit vs your project	Tag
yzhao062/pyod	[uncertain] / not verified here	Outlier detection toolkit.	active [uncertain]	Numeric/tabular arrays	Scores, labels, models	Strong anomaly layer for uncontracted issues.	[uncertain]
scikit-learn/scikit-learn	huge / current [uncertain]	General ML library including outlier and clustering methods.	active	Arrays/DataFrames	Estimators, predictions	Good primitives for isolation forest/LOF/One-Class SVM.	[uncertain]
sintel-dev/pyalibi-detect	[uncertain] / not verified here	Outlier, drift, and adversarial detection.	active [uncertain]	Arrays, images, text, tabular	Drift/outlier detectors and scores	Strong drift-detection backbone.	[uncertain]
online-ml/river	[uncertain] / not verified here	Online ML and drift detection.	active [uncertain]	Streaming data	Drift stats, online models	Strong for streaming or incremental DQ signals.	[uncertain]
evidentlyai/evidently	[uncertain] / not verified here	Monitoring and drift analysis for ML/data.	active [uncertain]	DataFrames, model outputs	Reports, JSON metrics [uncertain]	Good drift and quality monitoring, but report-heavy.	[uncertain]
datasketch/datasketch	[uncertain] / not verified here	MinHash/LSH sketches for similarity and near-dup detection.	active [uncertain]	Sets, strings, streams	Sketches, similarity estimates	Excellent for near-duplicate detection and scalability.	[uncertain]

ADJACENT — LLM-as-advisor over structured findings

Repo	Stars / date observed	Description	Status	Inputs	Outputs	Fit vs your project	Tag
medium/example	n/a	Prompt-based rule generation from samples.	n/a	Sample data, prompts	JSON rules / SQL text	Illustrates the pattern, but not a robust implementation.	[uncertain] [12]
PandasAI	[uncertain] / not verified here	LLM assistant for dataframe queries and analysis.	active [uncertain]	DataFrames, prompts	Text answers, code, charts [uncertain]	Adjacent only; too narrative/chat-oriented.	[uncertain]

Repo	Stars / date observed	Description	Status	Inputs	Outputs	Fit vs your project	Tag
LIDA	[uncertain] / not verified here	LLM-based data visualization assistant.	active [uncertain]	DataFrames, prompts	Charts, summaries	Useful contrast, but not your target architecture.	[uncertain]
D-Tale	[uncertain] / not verified here	Interactive dataframe exploration UI.	active [uncertain]	DataFrames	Web UI, plots, stats	Useful contrast for EDA/UI, not contract-driven DQ.	[uncertain]

Commercial Tools

Vendor / product	Description	Contract-aware overlap	Pricing tier / note
Monte Carlo	Data observability platform focused on anomaly detection and lineage.	Strong on monitoring and root cause; contract input is not the core public story.	Enterprise pricing; not publicly transparent [uncertain].
Anomalo	Automated data quality and anomaly detection platform.	Strong on statistical DQ, weak public evidence of declarative contract ingestion.	Enterprise pricing [uncertain].
Acceldata	Data observability and pipeline monitoring platform.	Strong on observability; can fit structured findings but not a DBML-first contract model.	Enterprise pricing [uncertain].
Bigeye	Data observability / quality monitoring.	Strong checks and monitoring; contract-driven input appears secondary.	Enterprise pricing [uncertain].
Lightup	Automated data quality and observability.	Good for anomaly detection and alerting; less visible contract-spec alignment.	Enterprise pricing [uncertain].
Telmai	Data quality and observability, especially for pipelines.	Overlaps with profiling and anomaly alerts, not a declarative schema contract core.	Enterprise pricing [uncertain].
Soda Cloud / Soda	Checks, monitoring, and observability around data sources.	Closer than most to rule-based DQ contracts; YAML-based checks are conceptually adjacent.	Free/OSS core + paid cloud tiers [uncertain].
IBM InfoSphere QualityStage	Enterprise data quality and matching/standardization suite.	Strong rule-based quality workflows; IBM also exposes data contracts in docs.	Enterprise licensing. [2]
Informatica Data Quality	Enterprise DQ suite with cleansing, matching, validation.	Strong governance and rules, but contract-first DBML-like usage is not evident.	Enterprise licensing.
Collibra Data Quality	Governance-oriented DQ.	Good governance integration; contract-specific public evidence limited.	Enterprise pricing [uncertain].
DataKitchen Enterprise	DataOps and testing platform.	Closest commercial angle for pipeline test automation and operational checks.	Enterprise pricing [uncertain].
Hex	Analyst workspace and AI assistant.	Adjacent only; useful contrast for analyst-assistant positioning, not DQ core.	Freemium/enterprise [uncertain].
Tableau Pulse	BI assistant/narrative layer.	Very distant from your architecture; helpful only as contrast.	Included in Tableau licensing. [uncertain]

Gap Analysis

- Well-covered capabilities.** Existing tools cover schema validation, row-level rule checks, profiling statistics, drift detection, and outlier detection quite well. Great Expectations, Frictionless, dbt tests, Soda, Deequ, TFDV, and profiling libraries already implement much of the “ML/statistical engine → findings” portion, even if they differ in input style and output shape. [\[6\]](#) [\[8\]](#) [\[9\]](#) [\[10\]](#)
- Declarative schema as contract.** This pattern is strong for Frictionless Table Schema, Great Expectations expectation suites, dbt schema tests/contracts, and IBM-style data contracts. It is weaker in pure profiling

libraries and in many observability platforms, where the contract is either implicit, SQL-based, or not central to the user experience.^{[2] [8] [7] [6]}

3. **DBML as DQ contract.** I did not find a credible public example of DBML being used specifically as a data-quality contract; the closest analogues are Frictionless Table Schema, Great Expectations suites, dbt contracts/tests, and IBM data contracts. The gap is therefore not just a missing implementation detail but an absent ecosystem pattern: DBML is schema-centric, while DQ-contract systems are usually validation-centric, so the project would be defining a new bridge between schema design and validation semantics.^{[8] [7] [10] [2]}
4. **Weakly covered or missing.** The biggest gaps are: one pipeline that accepts DBML plus live DB introspection plus files; simultaneous schema-vs-data mismatch and uncontracted anomaly detection; a deliberately versionable findings schema; and an LLM that advises on both data fixes and contract changes without doing the statistics itself. Existing tools usually do one or two of these, but not the whole chain in one architecture.^{[3] [9] [10] [6] [8]}
5. **Output-schema conventions.** Great Expectations gives rich JSON-serializable validation results; Frictionless gives compact validation reports; dbt gives test artifacts; Deequ yields constraint and metric outputs; OpenLineage facets are lineage-oriented rather than DQ-first; Soda emits scan results. As a baseline, Great Expectations' JSON result structure is the best starting point for extension because it already separates check-level evidence, aggregate status, and metadata.^{[7] [10] [6] [8]}
6. **Best differentiator.** The strongest differentiator is a contract-native, machine-readable DQ artifact that unifies schema mismatch, statistical profiling, and anomaly detection in a single findings file, then passes only those findings to an advisory LLM. That is different from narrative BI assistants and different from pure validation frameworks because the system stays ML-first and report-light.^{[3] [6] [8]}
7. **Technical risks.** The unresolved risks are LLM hallucination in remediation advice, false positives in drift/anomaly flags, performance on wide or high-cardinality tables, schema drift in semi-structured columns, safe execution of generated remediation code, and contract/data co-evolution over time. These risks are visible across the current ecosystem, but no single tool appears to solve them end-to-end.^{[6] [8] [3]}

Suggested Reading Order

1. **Profiling relational data: a survey** — best foundation for profiling tasks, metrics, and classic relational DQ methods.^[1]
2. **Frictionless validating-data docs** — clearest lightweight contract-driven validation model for tabular data.^[8]
3. **Great Expectations Expectation Suite + validation result docs** — strongest structured-output precedent for a findings file.^{[7] [6]}
4. **IBM data contracts docs** — practical example of declarative contracts executed as DQ tests and stored in Git.^[2]
5. **Quality Assessment of Tabular Data using LLMs** — nearest precedent for a modern “statistical engine + LLM rule/help layer” pipeline, useful mainly as a contrast point.^[3]



1. <https://uwaterloo.ca/data-systems-group/publications/profiling-relational-data-survey>
2. <https://www.ibm.com/docs/en/ws-and-kc?topic=quality-ensuring-data-data-contracts>
3. <https://www.arxiv.org/pdf/2509.10572v1.pdf>
4. <https://dl.acm.org/doi/10.1145/2566486.2568002>
5. <https://www.sciencedirect.com/science/article/pii/S2452414X24001729>
6. https://docs.greatexpectations.io/docs/reference/api/core/ExpectationSuiteValidationResult_class
7. https://docs.greatexpectations.io/docs/0.18/reference/learn/terms/expectation_suite
8. <https://framework.frictionlessdata.io/docs/guides/validating-data.html>
9. <https://github.com/frictionlessdata/frictionless-py>
10. <https://github.com/calogica/dbt-expectations>
11. <https://github.com/orgs/great-expectations/repositories>
12. <https://medium.com/@dabodie/automate-data-quality-with-an-llm-17db76049187>
13. <https://oscarpubhouse.com/index.php/sdlajast/article/view/41>
14. <https://www.acceldata.io/blog/how-ai-is-transforming-data-quality-management>

15. <https://www.coursera.org/articles/ai-tools-to-improve-data-quality>
16. https://www.linkedin.com/posts/gaurav-patole-20532154_data-contracts-for-data-quality-ok-so-activity-7341735318284328960-LOCw
17. <https://www.dqlabs.ai/blog/data-quality-automation/>
18. <https://hpi.de/naumann/survey-on-data-profiling-published-in-vldb-journal.html>
19. <https://github.com/frictionlessdata/frictionless-py/issues/809>
20. <https://frictionlessdata.io/tag/validator/>
21. <https://stackoverflow.com/questions/71505305/use-great-expectations-to-validate-pandas-dataframe-with-existing-suite-json>
22. <https://dblp.org/rec/journals/vldb/AbedjanGN15.html>
23. <https://www.x-mol.com/paper/1224988948373749766>
24. https://github.com/ckevinhill/great_expectations
25. <https://github.com/frictionlessdata>
26. <https://github.com/metaplane/dbt-expectations/actions>
27. <https://github.com/great-expectations>
28. <https://carpentries-incubator.github.io/frictionless-data-agriculture/01-introduction/index.html>
29. <https://discourse.getdbt.com/t/new-package-dbt-expectations/1771>
30. <https://github.com/kbaafi/great-expectations-intro>